# Bartlomiej Sobieski

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## About me

I am a second-year Ph.D. student in Computer Science at the University of Warsaw, under the supervision of Prof. dr hab. inż. Przemysław Biecek. My research focuses on developing and utilizing state-of-the-art generative modeling techniques to construct novel explainability algorithms for computer vision models. My greatest passion lies in exploring advanced mathematical methodologies and translating them into scalable, systems-level practical use cases.

## Publications (selected)

1. **Bartlomiej Sobieski***, Jakub Grzywaczewski*, Karol Dobiczek*, Mateusz Wójcik, Tomasz Bartczak, Patryk Szatkowski, Przemysław Bombiński, Matthew Tivnan, Przemyslaw Biecek. Auditing Sybil: Explaining Deep Lung Cancer Risk Prediction Through Generative Interventional Attributions. *International Conference on Machine Learning* (ICML), 2026.

2. **Bartlomiej Sobieski***, Matthew Tivnan*, Yuang Wang, Siyeop Yoon, Pengfei Jin, Dufan Wu, Quanzheng Li, Przemyslaw Biecek. System-Embedded Diffusion Bridge Models. *Neural Information Processing Systems* (NeurIPS), 2025.

3. **Bartlomiej Sobieski**, Jakub Grzywaczewski, Bartlomiej Sadlej, Matthew Tivnan, Przemyslaw Biecek. Rethinking Visual Counterfactual Explanations Through Region Constraint. *International Conference on Learning Representations* (ICLR), 2025.

4. **Bartlomiej Sobieski**, Przemyslaw Biecek. Global Counterfactual Directions. *European Conference on Computer Vision* (ECCV), 2024.

5. Hubert Baniecki, **Bartlomiej Sobieski**, Patryk Szatkowski, Piotr Bombinski, Przemyslaw Biecek. Interpretable Machine Learning for Time-to-Event Prediction in Medicine and Healthcare. *Artificial Intelligence in Medicine*, 2025.

6. Hubert Baniecki, **Bartlomiej Sobieski**, Piotr Bombiński, Patryk Szatkowski, Przemyslaw Biecek. Hospital Length of Stay Prediction Based on Multi-Modal Data Towards Trustworthy Human-AI Collaboration in Radiomics. *International Conference on Artificial Intelligence in Medicine*, 2023.

7. Krzysztof Jankowski, **Bartlomiej Sobieski**, Mateusz Kwiatkowski, Jakub Szulc, Michal Janik, Hubert Baniecki, Przemyslaw Biecek. Red-Teaming Segment Anything Model. *Computer Vision and Pattern Recognition Conference Workshops*, 2024.

## Education

**Doctoral School of Exact and Natural Sciences, University of Warsaw**
Warsaw, Poland

*Ph.D. in **Computer Science*** — 2024 - 2028 *(expected)*

- Advisor: Prof. Dr. Hab. Inż. Przemyslaw Biecek
- Research area: generative modeling for explainability in computer vision

**Faculty of Mathematics and Information Science, Warsaw University of Technology**
Warsaw, Poland

*M.Sc. in **Data Science*** — 2022 - 2024

- GPA: 4.92/5.00, graduated with honors.

**Faculty of Mathematics and Information Science, Warsaw University of Technology**
Warsaw, Poland

*B.Sc. in **Mathematics and Data Analysis*** — 2019 - 2022

- GPA: 4.68/5.00, graduated with honors.

## Work experience

**Centre for Credible AI**
*AI Researcher, Focus Lead* 2025.09 - now

**Harvard Medical School & Massachusetts General Hospital**
*Visiting Researcher* 2026.02 - 2026.04

**MI2.ai**
*AI Researcher* 2021.10 - 2025.09

## Participation in research grants (researcher)

**Centre for Credible Artificial Intelligence**
*Foundation for Polish Science (FENG.02.01-IP.05-0058/24)* 2025.09 - now

**DeMeTeR: Interpreting Diffusion Models Through Representations**
*National Science Centre (2023/50/O/ST6/00301)* 2024.10 - now

**xLungs: Responsible Artificial Intelligence for Lung Diseases**
*The National Centre for Research and Development (2019/34/E/ST6/00052)* 2022.07 - 2025.06

**PINEAPPLE: Explainable AI for hyperspectral image analysis**
*European Space Agency (ESA AO/1-11524/22/I-DT)* 2024.07 - 2024.09

**Bio-inspired Artificial Neural Networks**
*Foundation for Polish Science (POIR.04.04.00-00-14DE/18-00)* 2023.03 - 2023.08

**HOMER: Human Oriented autoMated machinE leaRning**
*National Science Centre (2019/34/E/ST6/00052)* 2022.03 - 2022.06

## Research-related activities

- **Organizer**, Generative AI meets eXplainable AI Special Session
  The 3rd World Conference on eXplainable Artificial Intelligence 2025 2025.07
- **Organizer and Meta-reviewer**, 4th XAI4CV Workshop
  Computer Vision and Pattern Recognition Conference 2025 2025.06

## Awards and Honors

- **2nd Place**, I National Competition for the Best Theses In The Artificial Intelligence Domain, Quantum AI Foundation 2025.02
- **1st Place**, XLI National Competition for the Best Master's Thesis In Computer Science, Polish Information Processing Society 2024.12
- **Scholarship of the Minister of Science and Higher Education**, Significant Scientific Achievements 2023.10
- **Academic Scholarship**, Outstanding Academic Achievements, Warsaw University of Technology 2019.10 - 2024.10

## Skills

**Deep Learning & Frameworks**: `PyTorch`, `JAX`, `Python`. Extensive experience implementing, training, and fine-tuning modern machine learning models and foundation architectures.

**Distributed Systems & Open Source**: Multi-GPU distributed training (**DDP**, **PyTorch Lightning Fabric**), HPC cluster management, SLURM workload manager, configuration management (**Hydra**), and experiment tracking (**WandB**). Creator and maintainer of an open-source scalable PyTorch research template designed to streamline deep learning workflows.

**Agentic AI Workflows**: Proficient in utilizing AI-assisted coding systems (Copilot, Cursor) to rapidly accelerate research code development, streamline debugging, and construct complex evaluation pipelines.

**Mathematics**: In-depth understanding of advanced calculus, linear algebra, statistics, probability theory, differential equations, and differential geometry.